7. SLIDING WINDOW

Aim:-

To write a program to implemant flow control at data link y layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

program should achieve at least below give requirements. you can make it a bidirectional program wherein receiver is sending its data frames with acknowlegement (piggybacking).

Create a sender program with following features:

1). Input window size from the user.

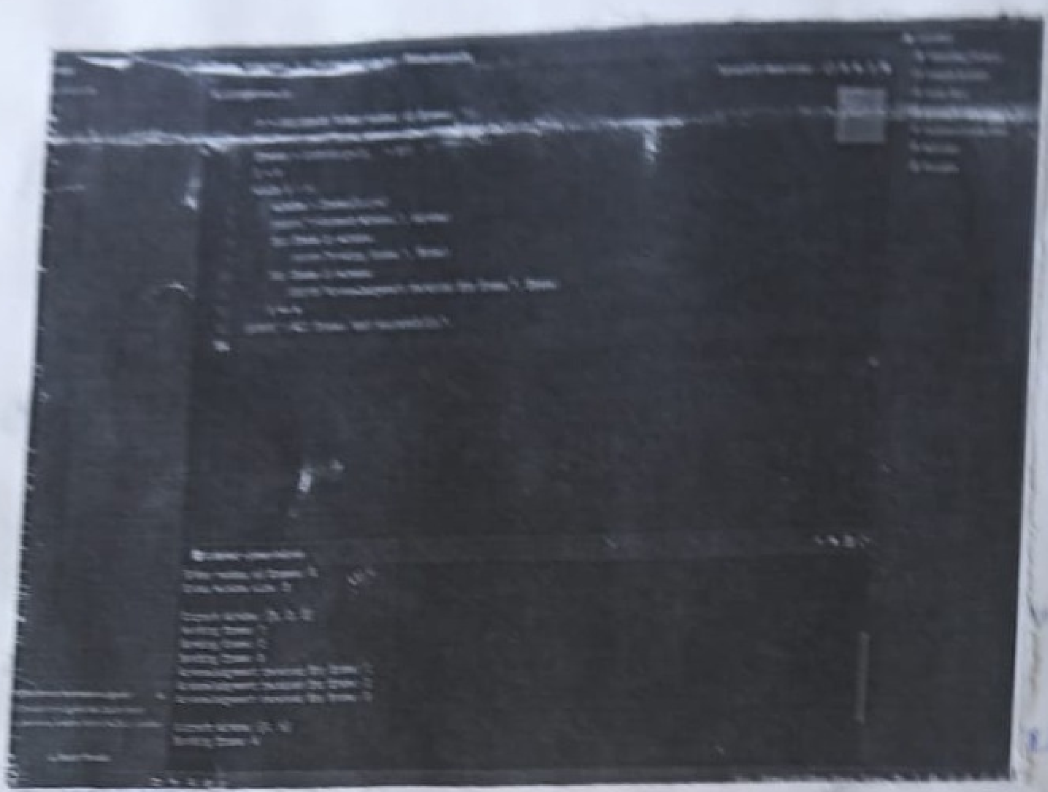
2) Input a Text message from the user.

3) Consider 1 character per frame.

4) create a frame with following fields [frame no. Data].

5) Send the frames. [print the output on screen and save it in a file called Sender-Buffer].

6) Wait for the acknowlegement from the Receive.



7) Reader a file called Reciever buffer.

8) Check ACK feild the Acknowlegement number

9) If the Acknowlegement number is as expected Send new set of frames accordingly.

Create a reciever file with following features:

1) Reader a file called Sender-Buffer

2) Check the Frame no.

3) If the Frame no are as expected. write the appropriate ACK no in the Reciever-Buffer file. Else write NACK no. in the Reciever-Buffer file.

Sender code:

```
import os.
import time
Base_DIR = os.path.dirname (os.path.abspath ( )
print ("Sender Buffer", Sender_File).
print ("Reciever Buffer", Reciever-File).
window_size = int (input ("Enter window size: ")).
message = input ("Enter the text message: ").
if window-size <= 0;
    print("window size must be greather than zero.")
    false systemed;
frames [].
while Start <total-frames:
    end = minstart + window-size, total-frames).
    print("\n = = = = = = = = = = = = = = = -")
    print("Sending ~~names~~ frames", Start + 1, "to", end)
    print(" = = = = = = = = = = = = = = = ")
  elif ack-type == "NACK"!
      print("NACK received")
      print("Retransmitting from frame:", ack_no)
   else:
      print("unknown response: Retrying current window:")
print("\n . . . . . ")
print("Total frames", total-frames):
```

Reciever code.

```
import os.
import time.
expected-frame = 1
last-content = ""
for line in lines
  try: parts = line . Split(";", 1)
  except (IndexError, valueerror.
      break;
```

```
else:
    print ("Frame error detected")
    print ("Expected frame:", expected_frame)
    error_found = True
print("\waiting for next window...")
time.sleep(0,5).
```

~~Input~~ .

Sender code Input

```
window size: 3
Message: HELLO
```

Sender code output

```
Frame 0: H
Frame 1: E
Frame 2: E
Frame 3: L
Frame 4: O
```

Receiver code Input.

Sender Buffer.txt

```
0  H
1  E
2  L
3  L
4  O.
```

Receiver code output

```
Received 0: H    Received 3: L
Received 1: E    Received 4: O.
Received 2: L
```

Result:

Therefore, the implement flow control of datalink layer using sliding window protocol executed successfully.